

Ren Lab/Center for Epigenomics: Single nucleus ATAC-seq protocol for CEMBA**A Nuclei preparation**

Preparation: Turn on the Thermal mixer to 37°C, turn on the swing bucket centrifuge and precool to 4°C.

1. Transfer nuclei suspension to Lobind tubes (Eppendorf) and spin down nuclei using a swinging bucket centrifuge (500 x g, 5 min, 4°C, Ramp rate: 3/9 acceleration & 3/9 deceleration; important: slow deceleration; 5920R, Eppendorf). Carefully remove and discard supernatant.
2. Resuspend pellets in 1 ml of NPB.

Composition of nuclei permeabilization buffer (NPB):

Reagent	Stock concentration	Final concentration	Amount per 10 ml
BSA (A7906, Sigma)	--	5 %	500 mg
IGEPAL-CA630 (I8896, Sigma)	10 % (m/V)	0.2 % (m/V)	200 µl
DTT (D9779, Sigma)	200 mM	1 mM	50 µl
cOmplete EDTA-free protease inhibitor (05056489001, Roche)	25x	1x	400 µl
PBS (10010-23, Thermo Fisher Scientific)	--	--	9.35 ml

3. Spin down nuclei using a swinging bucket centrifuge (500 x g, 5 min, 4°C, Ramp rate: 3/9 acceleration & 3/9 deceleration; important: slow deceleration; 5920R, Eppendorf). Carefully remove and discard supernatant.
4. Resuspend pellets in 500 µl of TB.

Composition of tagmentation buffer (TB)

Reagent	Stock concentration	Final concentration	Volume per 10 ml
Tris-acetate (pH=7.8, BP-152, Thermo Fisher Scientific)	1 M	33 mM	330 µl
K-acetate (P5708, Sigma)	3 M	66 mM	220 µl
Mg-acetate (M2545, Sigma)	300 mM	11 mM	367 µl
DMF (DX1730, EMD Millipore)	--	16 %	1.6 ml
Molecular biology water (46000-CM, Corning)	--	--	7.483 ml

5. Count with hemocytometer (4 μ l nuclei suspension, 6 μ l TB, 10 μ l Trypanblue). Count/field*50 is number of nuclei/ μ l in the suspension
6. Use volume representing 240,000 nuclei (120 wells at 2000 nuclei; please note: lower limit for experiment are 100,000 nuclei at this step, in this case reaction volumes need to be adjusted accordingly) and adjust volume to 1080 μ l --> 9 μ l per well (to 96 well plate). Keep plate and nuclei on ice.

B Tagmentation and sorting

Please note: If libraries for two samples are prepared on the same day in parallel, perform tagmentation in separate plates. Do not mix before sorting. Sort 20 nuclei from each sample into wells.

7. Add 1 μ l barcoded Tn5 transposomes (0.75 μ M, please note: optimal concentration needs to be tested for each new batch of Tn5) per well, e.g. using a BenchSmart™ 96 (Mettler Toledo) and mix 5 times.
8. Incubate at 37°C for 1h with 500 rpm shaking.
9. During tagmentation: Prepare sort plates e.g. using a Biomek i7 Automated Workstation (Beckman Coulter) with 7.5 μ l EB (Qiagen), 1 μ l barcoded primer i7 (12.5 μ M, IDT), 1 μ l uni primer i5 (12.5 μ M, IDT), 1 μ l BSA (200 ng/ μ l, 1:100 dilution of 2% BSA (A7906, Sigma) in PBS).
10. To inhibit the Tn5 reaction, add 10 μ l 40 mM EDTA (final 20 mM; 15575020, Thermo Fisher Scientific) to tagmented samples to quench the reaction and pipette 5x e.g. using a BenchSmart™ 96 (Mettler Toledo).
11. Incubate at 37°C for 15 min with 500 rpm shaking.
12. Add 20 μ l 2 x sort buffer (2 % BSA, 2 mM EDTA in PBS), mix 5x and combine in a reagent reservoir.
13. Stain nuclei suspension with 3 μ M Draq7 (1:100, 7406, Cell Signalling)
14. Sort 20 nuclei per well into 768 wells using a SH800 (Sony).

C Fragment release and amplification

15. Shortly spin down sort plates
16. Add to each well 1 μ l 0.2 % SDS (15553035, Thermo Fisher Scientific) and mix e.g. using a Biomek i7 Automated Workstation (Beckman Coulter).
17. Incubate at 55°C for 7 min with 500 rpm.
18. Add 1 μ l 12.5% Triton-X and mix e.g. using a Biomek i7 Automated Workstation (Beckman Coulter).
19. Add 12.5 μ l NEBNext 2x PCR MasterMix (M0541, NEB) and mix e.g. using a Biomek i7 Automated Workstation (Beckman Coulter).
20. Amplify fragments by PCR.

Temperature profile:

Step 1: 72°C for 5 min

Step 2: 98°C for 30 s

[Step 3: 98°C for 10 s

Step 4: 63°C for 30s

Step 5: 72°C for 1 min]

(Repeat steps 3-5 for a total number of 11 cycles)

Step 6: keep at 12°C.

D Sample Purification

21. Prepare 3mL 80% EtOH
22. Transfer samples to reservoir using 12-channel pipette (mix 5x prior to transfer). Combine wells from all 8 plates -> gives about 15 ml.
23. Transfer ~8ml to one 50 ml Falcon. Add 40 ml PB buffer (19066, Qiagen) including pH-indicator (1:2500) and 2 ml Na-Acetate (3M, pH=5.2) and mix.
24. Transfer on 8 MinElute columns (28004, Qiagen) including 3 ml Extension tube (19587, Qiagen) and apply vacuum.
25. Wash with 700ul PE (Qiagen) on vacuum station.
26. Spin down 1:30 min with 21,000xg.
27. Transfer column to 1.5ml LoBind tube (Eppendorf) and apply 20 µl EB (19086, Qiagen) and let stand at room temperature for 2 min.
28. Spin down for 1 min with 15,000xg.
29. Repeat previous two steps with another 20 µl EB (19086, Qiagen) to increase yield.
30. Combine in one 1.5 ml LoBind tube (Eppendorf) -> about 320 µl.
31. Use 160 µl for cleanup with SPRIselect beads (B23318, Beckman Coulter).
32. Add 40 µl EB buffer (Qiagen) to the sample, and add 110 µl SPRIselect beads (B23318, Beckman Coulter) and pipette 10x.
33. Incubate 5 min.
34. Separate on magnetic stand.
35. Transfer supernatant to a new tube (around 300 µl).
36. Add 190 ul SPRIselect beads (B23318, Beckman Coulter) and pipette 10x.
37. Incubate 5 min at room temperature.
38. Separate on magnetic stand
39. Discard supernatant.
40. Wash beads using 500 µl 80% EtOH for 30s (repeat).
41. Dry beads for 5 min at room temperature.
42. Resuspend beads in 20 µl EB, pipette 10x.
43. Transfer 17 µl to a LoBind tube (Eppendorf).
44. Add 33 µl EB buffer, 75 ul SPRIselect beads (B23318, Beckman Coulter) and pipette 10x.
45. Incubate 5 min.
46. Separate on magnetic stand.
47. Discard supernatant.
48. Wash beads using 500 µl 80% EtOH for 30s (repeat).
49. Dry beads for 3 min at room temperature.
50. Resuspend beads in 20 µl EB (Qiagen), pipette 10x and separate on the magnetic stand.
51. Transfer 17 µl to a LoBind tube (Eppendorf).

E Quality control and sequencing

52. Quantify final libraries using Qubit (1 µl/sample, Qubit dsDNA HS Assay Kit (Q32851), Thermo Fisher Scientific) and check library size distribution using 4200 TapeStation (High Sensitivity D1000 ScreenTape (5067-5584) and Reagents (5067-5583), Agilent Technologies).
53. Quantify final library using the PerfeCTa NGS Quantification Kit - Illumina (95154-500, Quantabio).
54. Sequence on a HiSeq4000 or NextSeq500 (Illumina, PE 50 + 10 + 12 + 50 (Read1 + Index1 + Index2 + Read2)).

Sequencing primer:

Primer ID	Sequence
Read 1 Sequencing Primer	GCG ATC GAG GAC GGC AGA TGT GTA TAA GAG ACA G
Read 2 Sequencing Primer	CAC CGT CTC CGC CTC AGA TGT GTA TAA GAG ACA G
Index 1 Sequencing Primer	CTG TCT CTT ATA CAC ATC TGA GGC GGA GAC GGT G
Index 2 Sequencing Primer (for HiSeq4000 and NextSeq500)	GCG TGG AGA CGC TGC CGA CGA